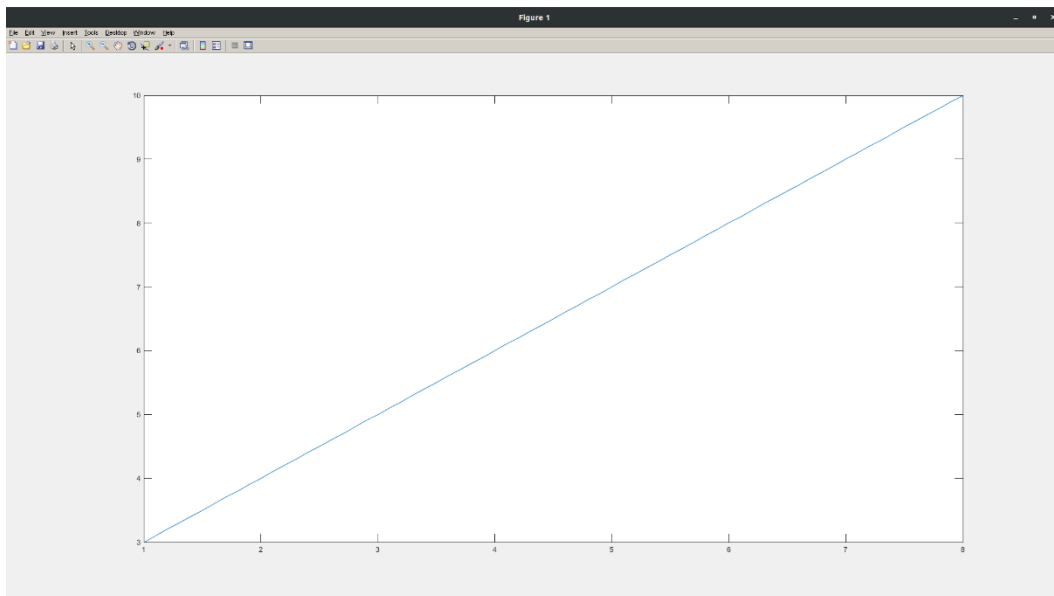


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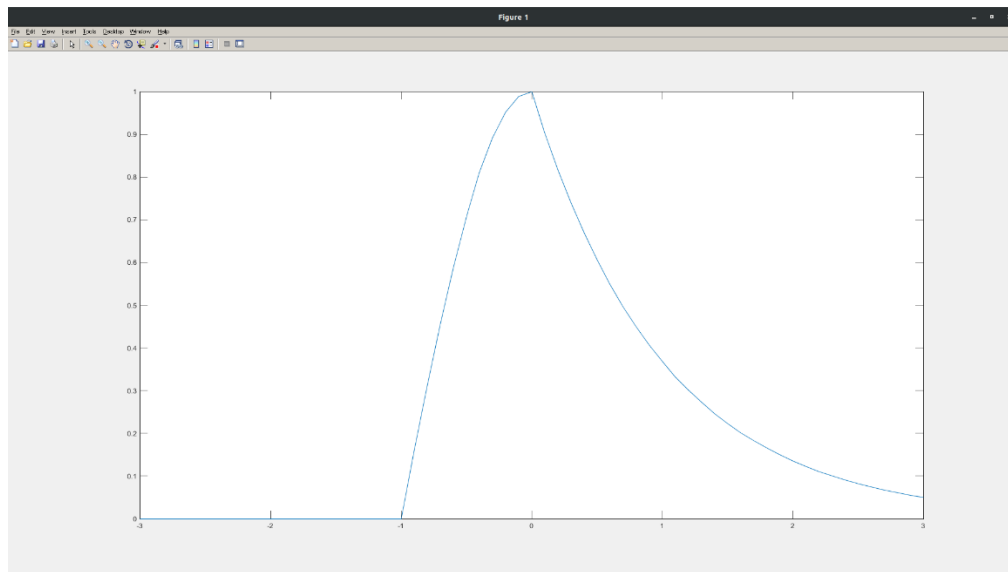
Άσκηση 1^η

```
t1=-10:0.1:-4;  
x1=0;  
plot(x1,t1)  
t2=-4:0.1:3;  
x2=t2+2;  
plot(x2,t2)  
t3=3:0.1:10;  
x3=t3-2;  
plot(x3,t3)
```



Άσκηση 2^η

```
t1=-3:0.1:-1;  
t2=-1:0.1:0;  
t3=0:0.1:3;  
x1=zeros(size(t1));  
x2=cos((pi*t2)/2);  
x3=exp(-t3);  
x= [x1 x2 x3];  
t= [t1 t2 t3];  
plot(t,x)
```

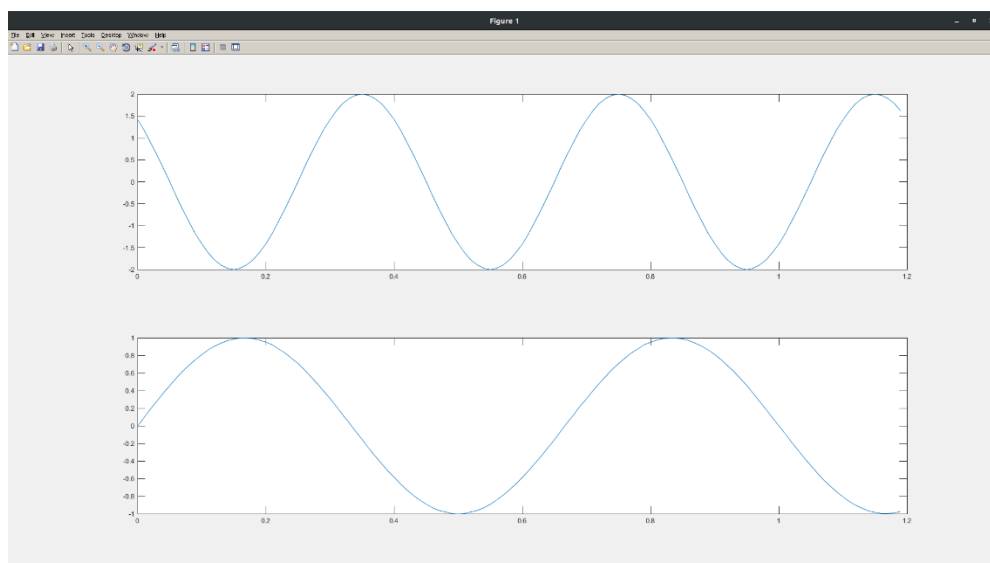


Άσκηση 3^η

```

T=2/5;
t=0:0.01:3*T-0.01;
x=2*cos(5*pi*t+pi/4);
subplot(2,1,1)
plot(t,x)
T1=2/3;
t1=0:0.01:3*T-0.01;
y=sin(3*pi*t1);
subplot(2,1,2)
plot(t1,y)

```



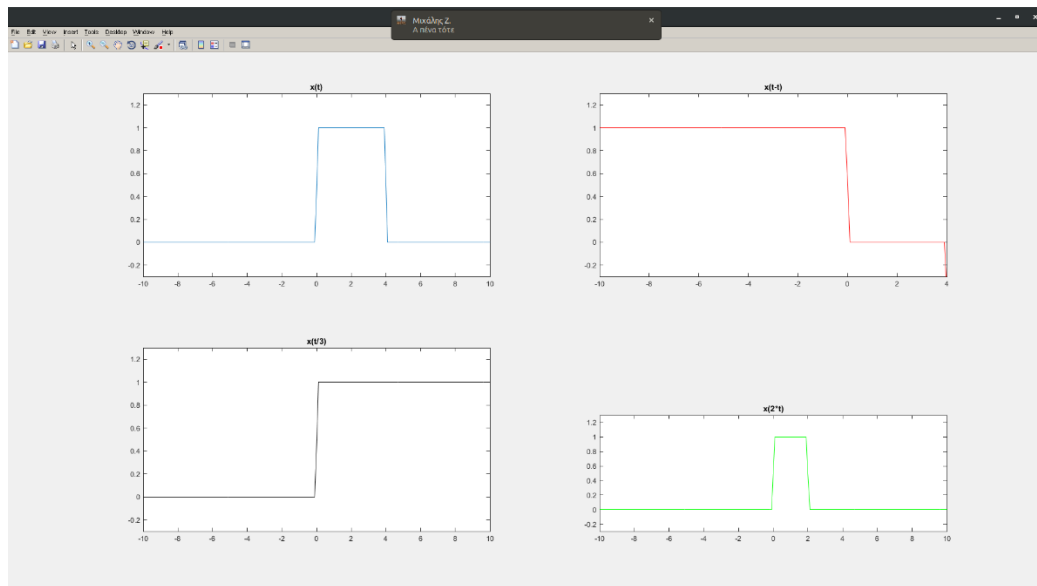
Άσκηση 5^η

```
subplot(2,2,1)
t=-10:.1:10;
u1=heaviside(t);
u2=heaviside(t-4);
u=u1-u2;
plot(t,u);
title('x(t)')
ylim([-0.3 1.3])
```

```
subplot(2,2,2)
u1=heaviside(-t);
u2=heaviside(t-4);
u=u1-u2;
plot(t,u,'r');
title('x(t-t)')
ylim([-0.3 1.3])
```

```
subplot(2,2,3)
u1=heaviside(t/3);
u2=heaviside(t/3-4);
u=u1-u2;
plot(t,u,'k');
title('x(t/3)')
ylim([-0.3 1.3])
```

```
subplot(2,2,4)
u1=heaviside(2*t);
u2=heaviside(2*t-4);
u=u1-u2;
subplot(3,2,6)
plot(t,u,'g');
title('x(2*t)')
ylim([-0.3 1.3])
```

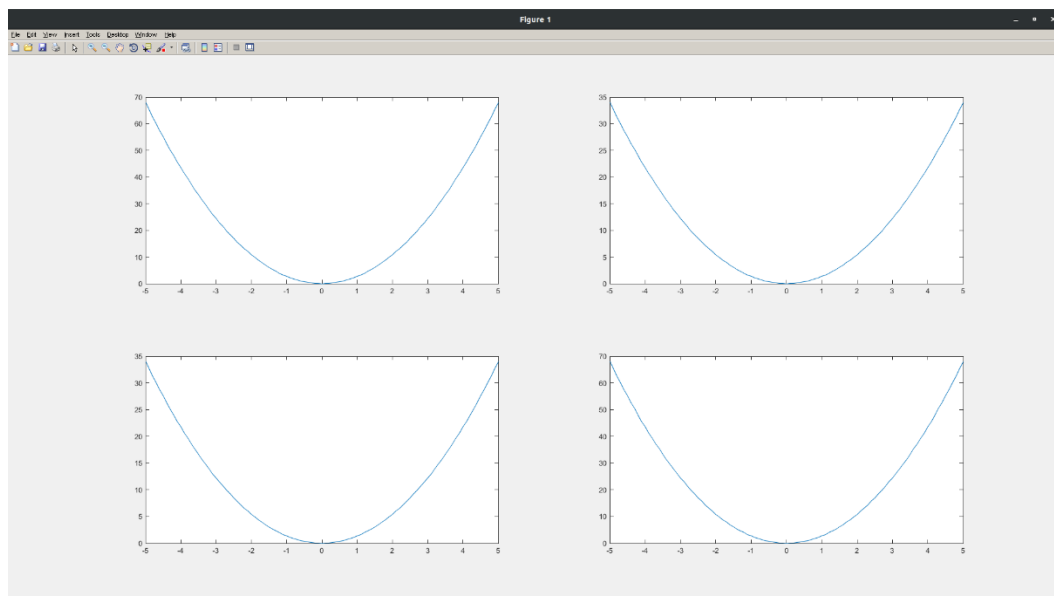


Άσκηση 6^η

```

t=-5:0.1:5;
x1=t.^2.*exp(1.^(3.*t));
subplot(2,2,1)
plot(t,x1)
x2= t.^2.*exp(1.^(3.*t))+(-t.^2.*exp(1.^(3.*(-t))))/2;
subplot(2,2,2)
plot(t,x2)
x3= t.^2.*exp(1.^(3.*t))-((-t).^2.*exp(1.^(3.*(-t))))/2;
subplot(2,2,3)
plot(t,x3)
x4=x2+x3;
subplot(2,2,4)
plot(t,x4)

```



Άσκηση 7^η

```
t=0:1:200;  
x1=rand(size(t));  
subplot(2,1,1)  
plot(t,x1),title('rand')  
x2=randn(size(t));  
subplot(2,1,2)  
plot(t,x2),title('randn')
```

